Non Perturbative Dynamics of Stochastic Inflation Beyond Slow-Roll

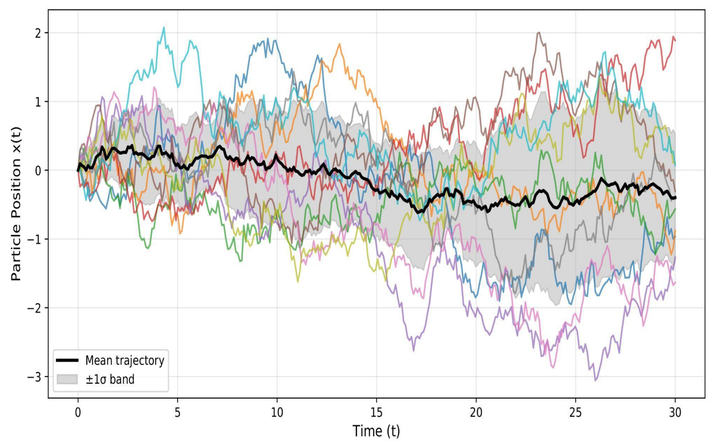
Devanshu Sharma[1]

Background

- Super-Hubble perturbations from inflation classicalize via Squeezing & Decoherence.
- Stochastic Inflation provides a bridge between quantum fluctuations and classicalized perturbations.
- We introduce a numerical algorithm to simulate the stochastic effects of the quantum fluctuations on de-Sitter expansion.
- We demonstrate this algorithm on different models and construct the two-point statistics of perturbations non-perturbatively using the ΔN formalism.

Brownian Motion

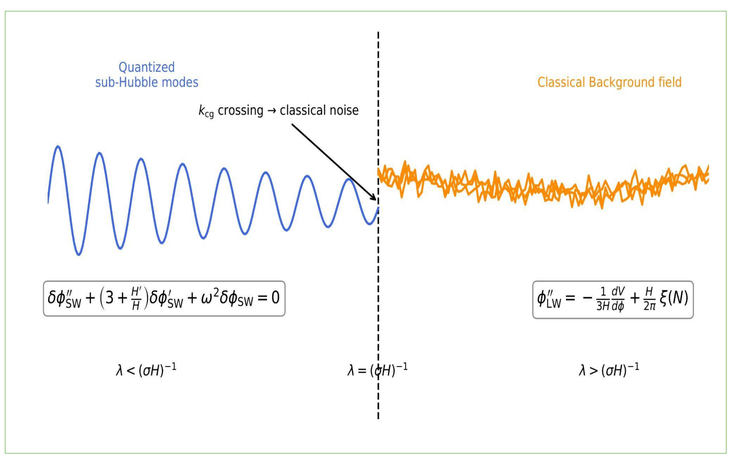
- Describes a stochastic process of motion of a particle suspended in a fluid.
- Equation of motion governed by Langevin Equation: Drift + Gaussian Noise.



- Each path represents an independent realization.
- Spread in the First Passage Time (FPT) crossing over multiple trajectories store the statistical moments (variance, skewness, kurtosis, etc.).

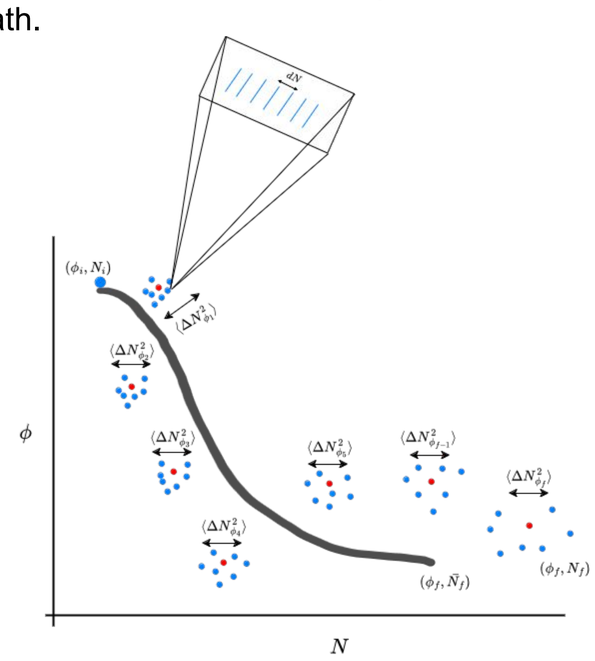
Stochastic Inflation (SI)

- Coarse-Graining splits inflaton into short-wavelength (SW) and long-wavelength (LW) modes.
- The Coarse-Grained inflaton (LW mode) trajectory draws analogy from the Brownian motion, with FPT being the end of inflation.



- Coarse-Grain Crossing: SW modes "kick" the background & undergo quantum to classical transition.

- The coarse-graining scale set sufficiently larger than the horizon to ensure the classicalization (squeezing parameter $r_k \gg 1$ for $\sigma = 0.01$).
- The kicks push the field away from its classical path.

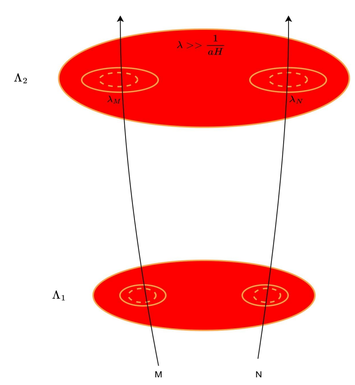


- Numerical implementation: Microscopic Trajectory Analysis with Stochastic Runge-Kutta method.

ΔN Formalism

- A non-perturbative technique to compute the curvature perturbations at super-Hubble scales.
- Separate Universe approach: Each point at large scales is an independent FLRW patch.
- Difference in the expansion rate between each FLRW patch gives the curvature perturbation:

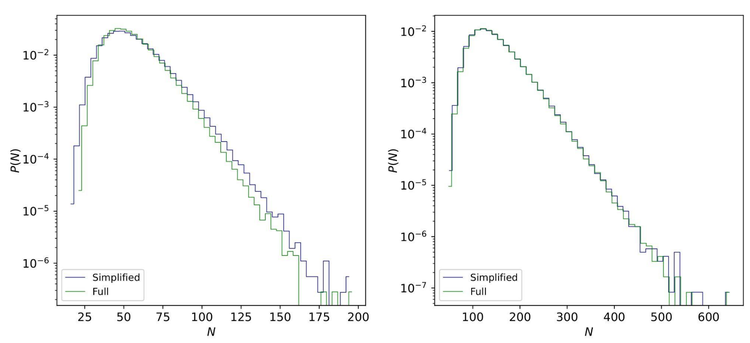
$$\mathcal{R} \equiv \Delta N$$



- Advantage: Captures non-linearities critical for Primordial Black Hole (PBH) formation.

Probability Distribution (PDF)

- First check: Toy model with an attractor regime.

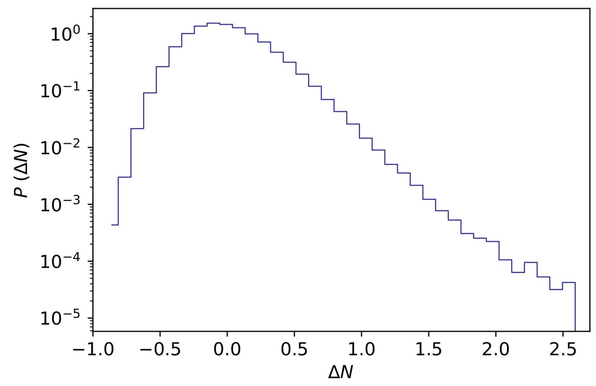


- Simplified Treatment: noise variance is equal to the constant super-Hubble solution:

$$\langle \xi_\phi^2 \rangle \sim \frac{H^2}{4\pi^2}$$

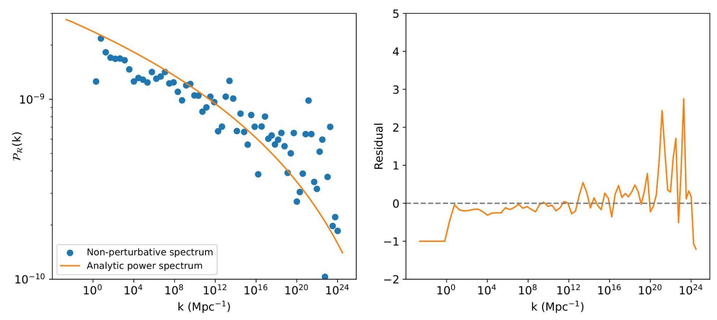
- Strongly non-Gaussian tails visible, courtesy of quantum diffusion.

- Second Check: High-energy theory motivated potential with an Ultra-Slow Roll (USR) regime.
- Important to resolve the tails deep inside the PDF (exponentially sensitive PBH abundance).



Power Spectrum

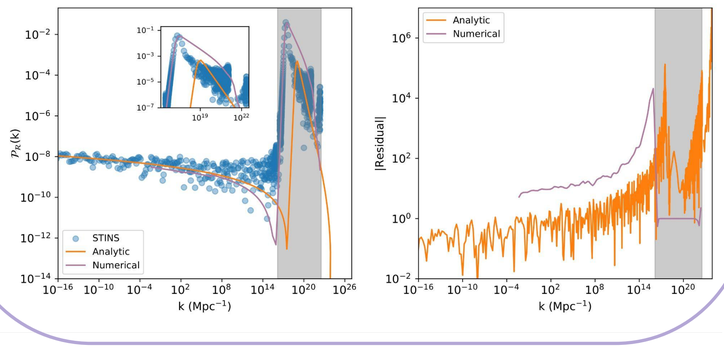
- Quadratic Potential: Recover the Slow-Roll (SR) power spectrum for the with good accuracy.



- USR Case: Work with "Super-hubble Time-Independent Non-perturbative Spectrum" (STINS).

$$\left.\frac{d\langle \Delta N_{\mathbf{k}}^2 \rangle}{d\bar{N}}\right|_{\bar{N}=\ln(k_f/k)} = \mathcal{P}_{\mathcal{R}}(k,\bar{N})\big|_{\bar{N}=\ln(k_f/k)} + \int_{k_i}^{k_f} \left(\frac{\partial \mathcal{P}_{\mathcal{R}}(k,\bar{N})}{\partial \bar{N}}\right) \frac{dk}{k}$$

- STINS framework: Recovers QFT power spectrum in attractor/non-attractor regimes.
- Residual spikes at the SR/USR transitions highlight the time-dependent effects.



Outlook

- Full non-perturbative numerical simulation applied for SR and USR models for the first time, using $\sim 4 \times 10^6$ stochastic realizations.
- Generic for single-field models; extensible to multi-field cases.
- Simulated using noise over a classical background; full stochastic analysis gives similar results. Work in progress.
- Higher-order moments (Bispectrum & Trispectrum) need more realizations to catch the tail behaviour. Work in progress.

References

1. DS, JCAP 03 (2025) 017, [arXiv:2411.08854 [astro-ph.CO]]

[1]Institute for Theoretical Particle Physics and Cosmology (TTK), RWTH Aachen University, D-52056 Aachen, Germany
(drsharma@physik.rwth-aachen.de)